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Low palaeolatitude of deposition for late Precambrian periglacial varvites in South Australia: implications for palaeoclimatology

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The natural remanent magnetisation (NRM) has been determined for periglacial clastic varvites (Elatina Formation) deposited in the Adelaide Geosyncline in South Australia during the Marinoan Glaciation ~ 680 m.y. ago. The varvites contain abundant silt-sized detrital specular titaniferous haematite and martite that experienced a high-temperature (~ 600°C) oxidation event prior to deposition, as well as ultrafine haematite and minor goethite pigment. The NRM intensities exhibited little change prior to thermal treatment above 600°C. The mean palaeomagnetic pole lies at latitude = 50.9°S, longitude = 337.1°E and $A_{95} = 2.2°$; the inferred palaeolatitude of the sampling locality is 5°. This pole position is similar to other late Precambrian poles for Australia. In view of the nature of the NRM and the lack of metamorphism of the rocks the magnetic remanence is regarded as primary (early diagenetic) and dating closely from the time of deposition. The contradiction of very cold climate near sea level in low palaeolatitudes during the late Precambrian brings into question basic tenets in palaeomagnetism and celestial mechanics.

1. Introduction

Late Precambrian glaciation ranks with the Mesozoic warm era as one of the outstanding climatic events in Earth history [1]. At times between 800 and 600 m.y. ago all continents with the possible exception of Antarctica experienced glaciation [1–5] during what has come to be regarded as "a climatic event of intensity unparalleled in the later history of the earth" [5].

The first suggestion that certain glaciogenic sequences of late Precambrian age accumulated in low palaeomagnetic latitudes [2,5–8] largely have been supported by subsequent palaeomagnetic investigations of glacial horizons [3,4,9–12]. Data for contiguous or penecontemporaneous non-glacial rocks generally are in agreement with this finding. Interestingly, a low ($\lesssim 30°$) palaeolatitude for glacial deposition is the preferred interpretation in most studies; there is little palaeomagnetic evidence from glaciogenic rocks for coeval high-latitude glaciation and attempts to relate glaciation to movement of continents through polar regions [13,14] have been unsuccessful. The glaciogenic deposits commonly are regarded as marine and may be associated with glaciated pavements and fossil permafrost horizons displaying periglacial structures such as sand wedges [3,4,15–17]. The surprising implication therefore is that grounded ice sheets and mean annual air temperatures below 0°C existed at or near sea level preferentially in low latitudes during the late Precambrian.

Some disagreement exists, however, over the interpretation of the palaeomagnetic data. The evidence for a low ($< 10°$) palaeolatitude of deposition for the Port Askaig Tillite in the Scottish Dalradian [18–20] is disputed by Stupavsky et al. [21], who argue that the tillite was remagnetised during the early Caledonian Orogeny in Ordovician time. The glaciogenic Rapitan Group of Canada revealed three remanence directions, two of which are regarded as Precambrian in age [22]; the simplest explanation is that the primary magnetisation was acquired in low (14°) palaeolatitudes, although other explanations are possible. Furthermore, Piper [23] queried numerous independent conclusions of low palaeolatitudes of de-

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position for late Precambrian glacial rocks, claiming that the results "are derived from drab sedimentary rocks, and in no instance can the magnetisation interpreted as primary be attributed with certainty to the age of deposition". McWilliams and McElhinny [10], on the other hand, in noting that a low latitude for glaciation is the preferred interpretation in all the studies they discussed, concluded "that the consistency of the available data argues for the validity of the low latitude interpretation". Nonetheless, reasonable doubts remain concerning the interpretation of certain palaeomagnetic data for late Precambrian glacial horizons.

Here we present new palaeomagnetic results from South Australia that bear on this controversy. The remanent magnetisation has been determined for the Elatina Formation, a unit of red siltstone and fine-grained sandstone locally displaying laminae interpreted as varves, that was deposited in a periglacial lacustrine environment bordering the Adelaide Geosyncline during the Marinoan Glaciation about 680 m.y. ago (Fig. 1). The Precambrian history of the Adelaide Geosyncline is discussed by Rutland et al. [24]. The varvites at Pichi Richi Pass in the Flinders Ranges record striking climatic periodicities of apparent solar origin [25–27] and they have been studied palaeomagnetically by us to gain a better understanding of the local geomagnetic environment at that time. The varvites exhibit a clear remanent magnetisation signature that is demonstrably primary and which indicates deposition near the palaeoequator. The results are in agreement with palaeomagnetic data for other late Precambrian rocks in Australia and provide perhaps the best evidence to date for a low palaeolatitude of deposition for late Precambrian glaciogenic rocks.

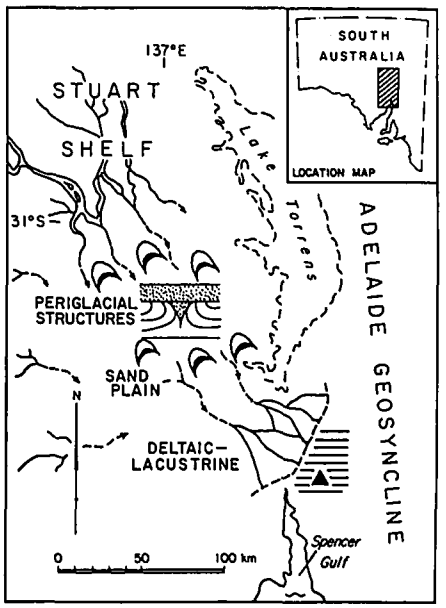


Fig. 1. Palaeogeographic scenario for north-central South Australia during the Marinoan Glaciation about 680 m.y. ago, showing: the sand plain with schematic barchan dune forms (Whyalla Sandstone); the location of the Marinoan fossil permafrost horizon buried by the Whyalla Sandstone; and the periglacial lake within which the varvites of the Elatina Formation accumulated (the triangle marks the sample site in Pichi Richi Pass). The source of the annual meltwaters lay to the west and northwest.

2. Sedimentary features

Samples for palaeomagnetic study were obtained from cored vertical drill holes and from outcrops of a 10-m-thick laminated (varvite) member within the Elatina Formation at Pichi Richi Pass (latitude 32°25'S, longitude 137°59'E). The member consists mainly of graded laminae ~ 0.2–3 mm thick of clayey siltstone to very fine sandstone (Fig. 2) which are attributable to annual deposition by density underflows in the distal portion of a periglacial lake [25]. Scattered "drop grains" up to the size of small pebbles probably were derived from melting river ice. The member is characterized by conspicuous groups or cycles of about 10 to 14 laminae that usually are bounded by thinner, darker, more clayey laminae. The thickness of graded laminae varies systematically within each cycle, attaining a maximum near the cycle centre. Longer rhythms are evident through regular changes in cycle thickness.

The laminae are interpreted as annual increments, i.e. varves, because of (1) their similarity to certain modern intermediate-distal varve facies [28–30], and (2) the independent evidence, provided by a penecontemporaneous fossil permafrost horizon in the palaeocatchment area, for a strongly seasonal climate during deposition of the Elatina Formation [17]. The varvite member spans an interval of time of about 2×10^4 varve years [26,27]. The cyclicity displayed by variations in varve and varve-cycle thicknesses is ascribed a solar-climatic origin [25–27].

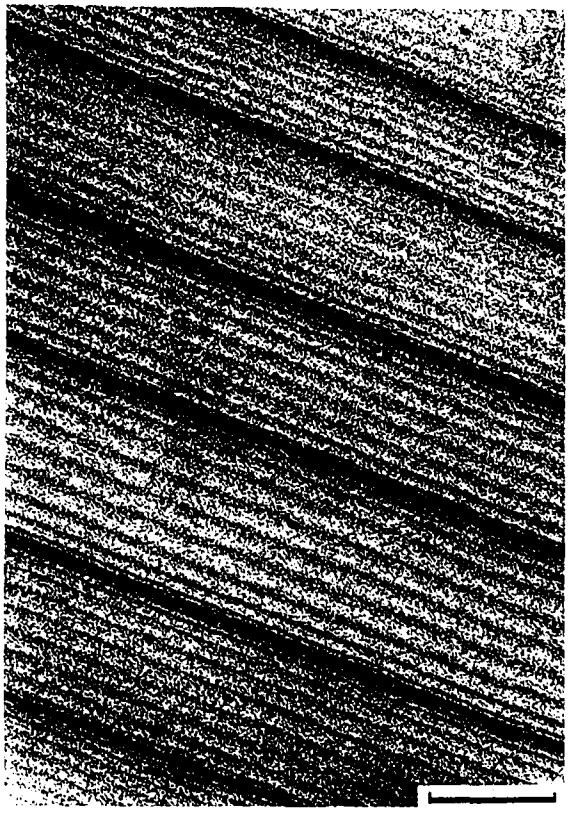


Fig. 2. Varvite member of the Elatina Formation from Pichi Richi Pass. Graded laminae of clayey silt to very fine sand grade are interpreted as varves. Conspicuous groups or cycles of about 10 to 14 varves that usually are bounded by thinner, darker, more clayey varves characterize the member. Scale bar 1 cm.

The sediments are unmetamorphosed, apart from normal burial metamorphism. This conclusion is based in part on the following observations [31]:

(1) Details of sedimentary structures such as varves, graded bedding, and heavy mineral banding are clearly and faithfully preserved.

(2) The original clastic outlines of individual grains are sharply defined, with minimal recrystallization of marginal corrosion.

(3) Scattered diagenetic pyrite nodules (largely oxidized to goethite) have irregular shapes and outlines; there is no evidence of recrystallization to euhedral forms, nor that pyrrhotite was produced.

(4) The only component that is recrystallized is the clay matrix, which is now fine, random sericite. Furthermore, there is no evidence of recrystallization of any of the detrital micas present. As the rocks contain neither authigenic muscovite nor chlorite, they cannot have been subjected even to the lower limit of stability of the greenschists assemblage, which may be placed at around $T = 300°C$ and $P_{H_2O} = 2$ kbar [32].

The lack of metamorphism of these rocks accords with their occurrence in the zone of least alteration within the Adelaide Geosyncline, defined by the presence of "subgraphitic" kerogen and poorly crystalline illite in sedimentary rocks stratigraphically well below (some 100 m.y. older than) the Elatina Formation [33]. The subgraphitic kerogen in this area (atomic $H/C = 0.11–0.19$, $C = 89–91\%$ dry, ash free) corresponds to a coal rank in the low volatile bituminous to semi-anthracite range (vitrinite reflectance = 1.5–2.0%). This indicates diagenetic temperatures no greater than 135–160°C, assuming an effective heating time (the time that the temperature of a given sediment has been within 15°C of the maximum temperature) of 250–300 m.y. [34,35].

3. Magnetic mineralogy

The varvite member is red in colour for unweathered material (Munsell colours 5R 5/1 (dry), 7.5R 3/2 (wet)) due to the presence of ultrafine haematite in interstitial clays and carbonate (?siderite) as well as minor goethite pigment. The rocks also contain abundant opaque silt-sized grains of Fe-Ti oxide. Two rock-chip specimens from the varvites assayed respectively 3.60% and 3.30% Fe_2O_3, and 0.64% and 0.26% FeO.

To obtain a representative concentration of the opaque heavy minerals from the varvites, approximately equal amounts from three core samples (diamond drill hole (DDH), 2, 56.98–57.10 m; DDH3, 4.70–4.90 m, 8.76–8.96 m) were combined, crushed to −36 mesh B.S.S., and separated in tetrabromoethane (specific gravity = 2.95); gravity settling proved ineffective and the material finally was centrifuged to obtain a concentrate, which was mounted and polished for petrographic study. The opaque phases identified are:

(1) Haematite with exsolution blades of rutile (Fig. 3a) or of both rutile and ilmenite, or with seriate exsolution bodies of ilmenite (Fig. 3b).

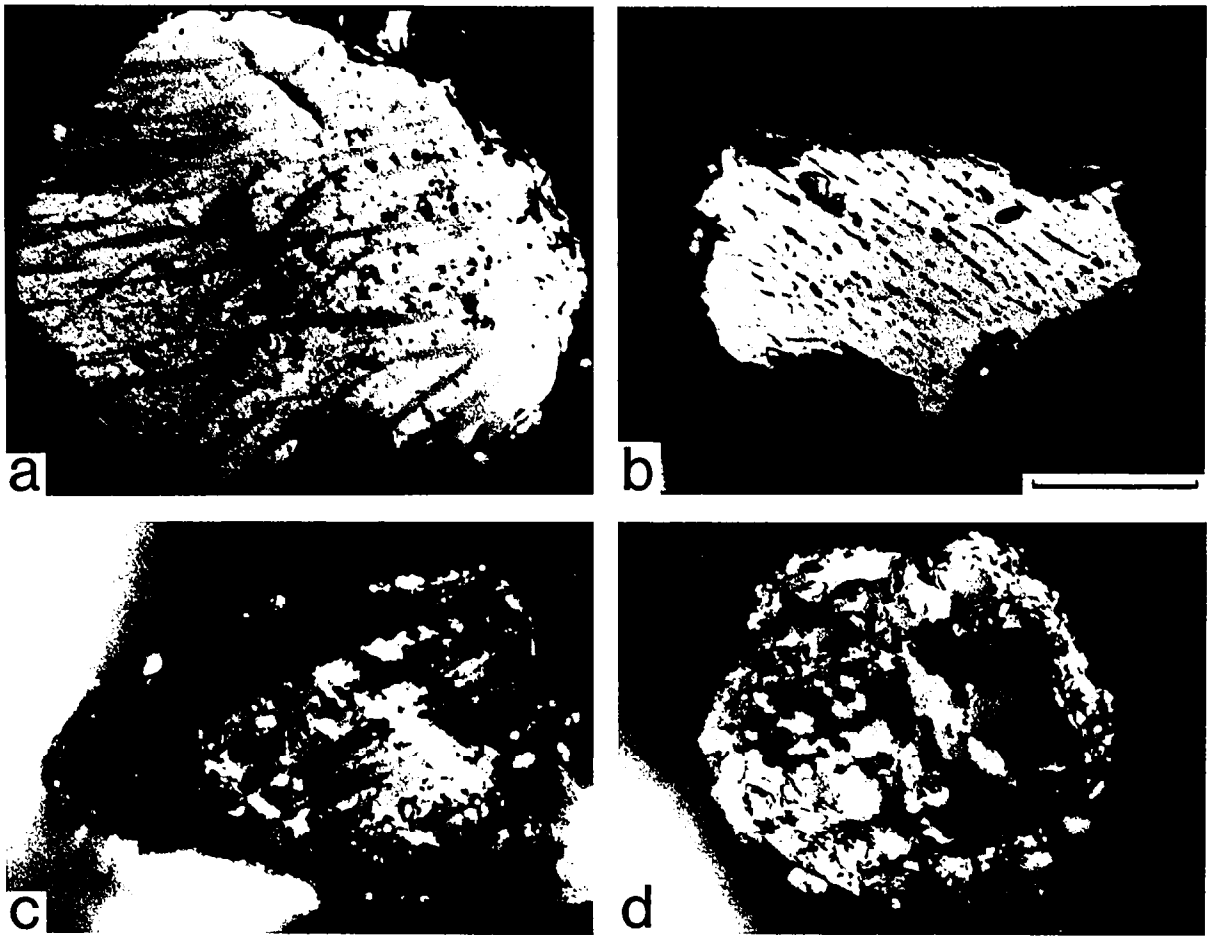


Fig. 3. Opaque heavy mineral grains from the varvite member. (a) High-temperature haematite (pale) with exsolution blades of rutile (b) Ti-haematite with seriate exsolution bodies of ilmenite. (c) Rounded detrital grain of martite-rutile intergrowth surrounded by rutile (crossed polars). (d) Martite showing a pattern of interlocking irregular patches (crossed polars). Scale bar 25 μm for all grains.

Such titaniferous haematite is of high-temperature (> 600°C) igneous or metamorphic origin [36–38], the exsolution textures forming by moderately slow cooling below about 600°C. As the Elatina Formation at Pichi Richi Pass clearly has not experienced such a high temperature, a detrital origin for these haematite grains is indicated.

(2) Martite (haematite) after magnetite, containing lamellae of rutile (Fig. 3c). The presence of rutile points to high-temperature (> 600°C) oxidation of the original magnetite [38]. A detrital origin for such martite grains is indicated. Groups 1 and 2 together constitute ~ 25% of opaque grains [31].

(3) Martite after magnetite, now entirely hematite but displaying a pattern of relic (111) octahedral planes of the former magnetite. Other martite grains show a pattern of interlocking irregular patches (Fig. 3d), some with relics of magnetite. The oxidation history of such grains is unclear but it is likely that some of the oxidation occurred after deposition. This group constitutes ~ 50% of opaque grains.

(4) Degraded ilmenite, generally comprising ultrafine intergrowths of "leucoxene" and haematite. Much of the breakdown of ilmenite probably occurred after deposition.

The petrographic evidence thus indicates that more than 30% of the haematite and martite grains experienced a high-temperature (> 600°C) oxida-

tion event prior to deposition of the Elatina Formation and have not been significantly altered by subsequent diagenetic processes. Such grains may confidently be regarded as detrital.

A suite of silt-sized opaque minerals similar to that in the varvites from dark heavy-mineral bands in immediately overlying cross-laminated sandstones (DDH1, 15.80 m). These heavy mineral bands are perceptibly magnetic, as material from drill core very slightly perturbs a pendulum-type field magnet.

4. Palaeomagnetism

The experimental work was conducted in two stages. First, drill cores were sampled at regular intervals from about 12 m of section spanning the varvite member, representing $\sim 2 \times 10^4$ varve years (17 samples from DDH1, 14.95–27.10 m, and 3 samples from DDH2, 55.60–57.60 m). The drill-core material was then sub-sampled and the natural remanent magnetisation (NRM) of 59 specimens was measured and tested for stability to alternating field and stepwise thermal demagnetisation. As the drill holes were vertical and outcropping varvites display a uniform westward dip at the drill site, the azimuthal orientation of the core samples was considered unambiguous.

Experimental procedures generally followed those described by Collinson et al. [39]. All measurements of magnetic remanence were carried out on a DIGICO Complete Results Magnetometer. A plot of 20 mean directions of drill-core NRM (each the mean of approximately three specimens per discrete sampling level in the core) prior to cleaning is shown in Fig. 4a. Following partial thermal demagnetisation up to the blocking temperature around 680°C (Fig. 4b), directions of magnetic remanence exhibited little change and retained the distinctive spread in declination though with only a small variation in inclination. The spherical median [40,41] for the cleaned directions is declination = 195°, inclination = −2.0° and the 95% elliptical confidence cone has semi-axes 7.6° and 2.0°. Attempts at alternating field demagnetisation failed to change the NRM intensities (generally around 20 mA m^{-1}) by more than a few per cent and again resulted in negligible direction changes. Susceptibilities were generally in the range 7 to 10×10^{-6} G Oe^{-1} ($\sim 100 \times 10^{-6}$ SI) and the susceptibility anisotropy (also measured on DIGICO equipment) in 20 specimens was consistently < 5%. Clearly the elliptical distribution of the NRM and cleaned remanence directions is due to imprecise azimuthal control through using the attitude of outcropping strata to orient the vertical drill core.

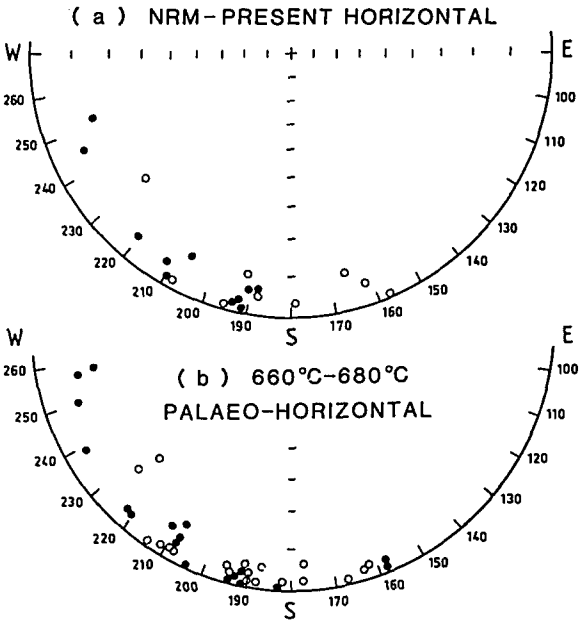


Fig. 4. Sample-average remanence directions shown in (a) for drill-core of the varvite member. Following thermal demagnetisation (b) individual specimen directions exhibit little directional change. The spread of declination values is attributed to the down-hole variation in strike when only a single strike measurement is available from outcrop.

The distinct clarity of the palaeomagnetic signature was, however, sufficiently diagnostic to encourage a full analysis based on a conventional palaeomagnetic collection. Surface samples were collected at six sites with a hand-held power drill and oriented using both sun and magnetic compass techniques. Limited outcrop over approximately 300 m restricted sampling to the west-dipping limb and south-plunging nose (axis) of a small anticline.

NRM directions before cleaning are remarkably uniform both within each site and between sites (Fig. 5). NRM intensities range from 10 to 40 mA m^{-1}. The magnetic signature is genuinely non-complex. Detailed thermal demagnetisation failed to reveal more than one significant compo-

TABLE 1

Summary of palaeomagnetic directions measured in surface samples of the laminated facies (Elatina Formation)

Site	Bedding attitude		With respect to horizontal		Bedding corrected		N	R	α
	strike	dip	dec.	inc.	dec.	inc.			
PR04	303 (fold axis plunge)	14 S [a]	192.5	+3.1	192.2	+10.0	5	4.997	1.9
PR05	333	17 SW	195.2	+7.1	194.4	−4.3	9	8.992	1.5
PR06	353	21.5 W	191.0	+3.4	188.7	−5.9	6	5.989	3.1
PR07	273 (fold axis plunge)	13 S [a]	191.5	+0.7	191.6	−12.1	5	4.961	7.5
PR08	328	14 SW	195.2	−1.4	192.7	−11.5	7	6.946	5.7
PR09	358	33 W	199.5	+1.7	192.4	−13.9	5	4.977	5.8
Mean			194.1	2.4			6	5.986	3.5
					191.9	−9.6	6	5.987	3.4

[a] Plunge measured at site 04 (07) used to correct magnetisation direction and bedding attitudes at nearby sites 05 and 06 (08 and 09).

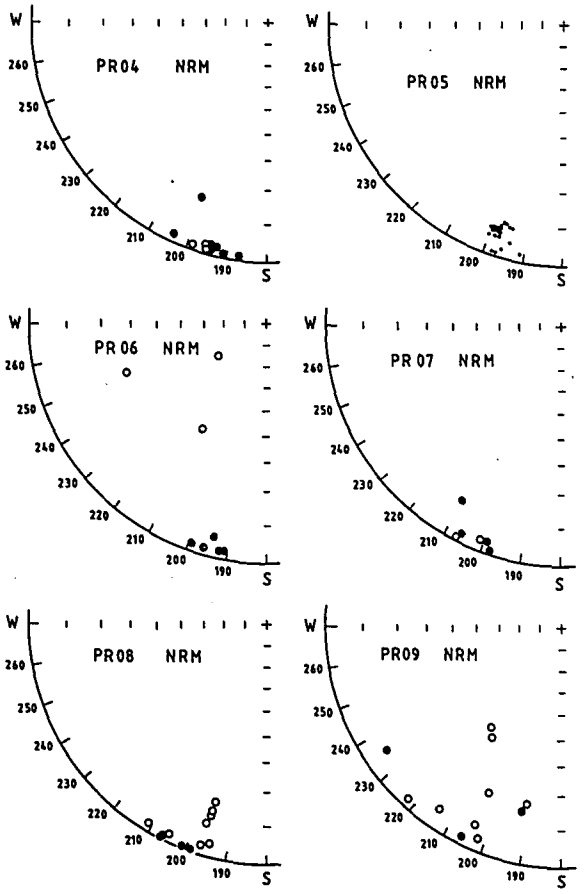


Fig. 5. Directions of natural remanent magnetisation at each of six sites measured from conventional palaeomagnetic specimens collected from outcrop.

nent of magnetisation. Orthogonal plots of the end-points of the magnetisation vectors for one specimen from each site are shown in Fig. 6. The extreme stability of the magnetisation is evident with the blocking temperature occurring close to the Curie points for haematite (around 680°C).

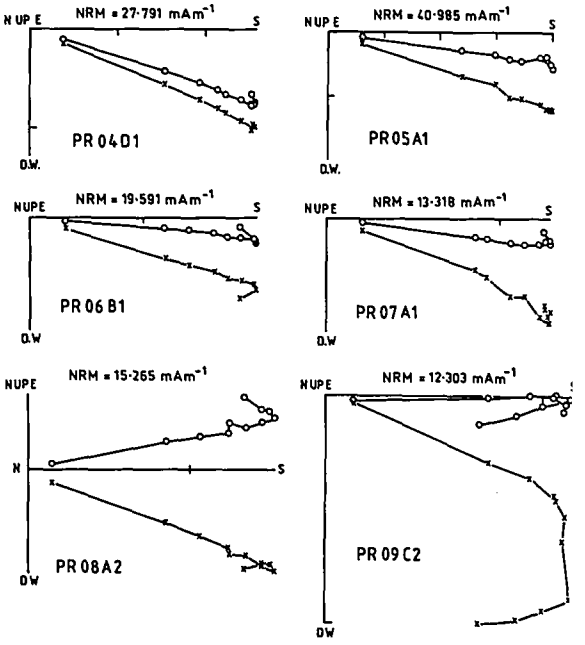


Fig. 6. Orthogonal plots of vector end-points acquired from experimental analysis of blocking temperature spectra using step-wise thermal demagnetisation procedures. Temperature intervals following NRM measurements were generally 200°, 300°, 400°, 500°, 600°, 630°, 640°, 660°, 670° and 680°C.

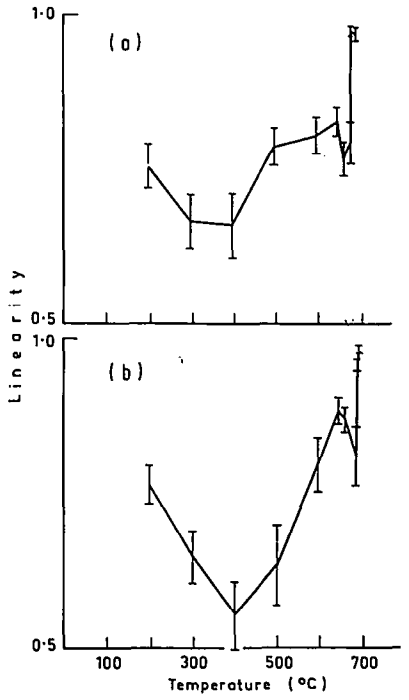


Fig. 7. Linearity spectrum graphs for two sub-sets of thermal demagnetisation data.

The analysis of the demagnetisation data was carried out using the linearity spectrum method [42]. The technique is particularly appropriate for this collection because of the evident uniformity of magnetic mineralogy throughout the entire section of the varvite member. Demagnetisation of two-sub-sets of specimens yielded group-mean directions with respect to the present horizontal at the sampling locality of declination = 194.9°, inclination = 1.7°, $\alpha_{95} = 3.5°$ and declination = 193.6°, inclination = 2.5°, $\alpha_{95} = 3.0°$. There is considerable variation of the linearity at low temperatures as well as the linearity itself being generally low (Fig. 7). However, following the onset of demagnetisation the linearity value increases rapidly and approaches unity close to the blocking temperature.

Specimen results combined to provide site-means are listed in Table 1. Despite the high precision that has been possible with this collection, the directions of magnetisation are too closely sub-parallel with the fold axis to enable diagnostic application of the fold test. Bedding-corrected results for each site (Table 1) give the following overall mean: declination = 191.9°, inclination = −9.6° and $\alpha_{95} = 3.4°$. The corresponding palaeomagnetic pole lies at latitude = 50.9°S, longitude = 337.1°E and $A_{95} = 2.2°$. The inferred palaeolatitude of the sampling locality is then 5°.

5. Discussion of palaeomagnetic data

The specular, silt-sized detrital haematite and the ultrafine haematite located in the interstitial clays contribute to a single, ubiquitous component of magnetisation. The higher blocking temperature is uniformly discrete and the demagnetisation behaviour may justifiably be described as "square shouldered." The NRM intensities exhibited little change prior to thermal treatment above 600°C. Dunlop and Buchan's [43] thermal unblocking model indicates that to remagnetise or thermally overprint a mineralogy yielding such stable magnetic behaviour, the rocks would require metamorphism under conditions appropriate to at least upper amphibolite facies (i.e. $\geqslant$ 600°C) [32]. As demonstrated in section 2, however, these rocks have been subjected to no more than burial metamorphism. The homogeneity of the data as indicated by (1) the absence of a spectrum of blocking temperatures, and (2) the absence of a multi-component NRM comprising overprint magnetic signatures is consistent with a relatively short period of acquisition of remanent magnetisation and presumably one effective only during early diagenesis. This is also the most likely time at which the direction of remanence was finally set for the detrital, unaltered grains.

If the magnetisation had been acquired over a protracted period of diagenesis then we may expect to have encountered considerably greater variation of magnetic properties, reflecting a distribution of grain sizes, and hence magnetisation stability. Furthermore, the rocks studied are unmetamorphosed and have not experienced temperatures anywhere near the determined blocking temperature of about 680°C. The magnetic remanence must be regarded as primary (early diagenetic) and dating closely from the time of deposition.

The characteristic direction yields a palaeomagnetic pole position similar to other Australian late Precambrian poles from the Adelaide Geosyncline [9,10,44], the Amadeus Basin [44,45] and the

Northampton Block [46]. This direction is quite distinct from directions of magnetisation ascribed by Klootwijk [44] to magnetic overprinting acquired during the Delamerian Orogeny that affected the Adelaide Geosyncline in the Late Cambrian and Early Ordovician. The absence of a Delamerian overprint in the Elatina varves can be attributed to their extreme magnetic stability. Apparent polar wander with respect to Australia is well documented for the late Precambrian through the early Palaeozoic [46,47] and the pole obtained from the varvite member of the Elatina Formation is indistinguishable from the group of poles for the quasi-static interval 700–560 m.y. [46]. The Elatina Formation was deposited during this interval. The mean 700–560 m.y. pole, latitude = 43.4°S, longitude = 336.2°E and $A_{95} = 5.7°$, indicates a depositional palaeolatitude of about 12° for the Elatina Formation at Pichi Richi Pass; this is in good agreement with the inferred palaeolatitude of 5° estimated herein from the magnetisation of the varvite member.

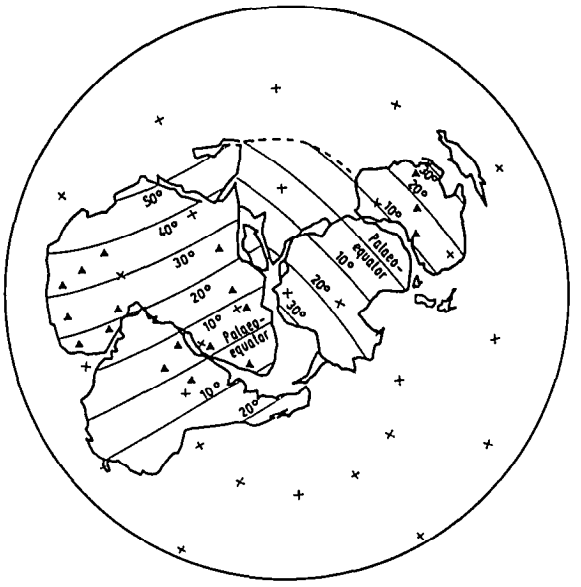


Fig. 8. Palaeolatitude models for east and west Gondwana for late Precambrian time based on data for glaciogenic deposits in Australia and southern Africa. The distribution of known occurrences of late Precambrian glaciogenic deposits is indicated (▲).

6. Wider implications

The palaeomagnetic data presented here, together with those for other late Precambrian glaciogenic rocks, place constraints on models for continental reconstruction and global climate during late Precambrian time.

A reconstruction of Gondwana [47], together with palaeolatitude models for east and west Gondwana based on data for late Precambrian glaciogenic rocks, are shown in Fig. 8. The palaeolatitude model for east Gondwana, based on Australian data, shows that the principal shield regions of India, Antarctica and Australia occupied low latitudes at that time. The palaeolatitude model for west Gondwana (South America, Africa, Arabia and Madagascar), based on data for the Nama Group of southern Africa [11], places the palaeoequator in southern Africa and the remainder of Africa in low to intermediate palaeolatitudes. Taken together, the two palaeolatitude models suggest either that (1) the shown reconstruction for Gondwana as a whole is invalid for late Precambrian glacial time [48], or (2) late Precambrian glaciation was non-synchronous despite its apparent preference for low palaeolatitudes. The palaeomagnetic results demonstrate the critical value of having azimuthal control to constrain palaeolatitude models.

The occurrence of ice sheets and periglacial climate near sea level in low palaeolatitudes is a major enigma. As continental glaciation is normally associated with high latitudes and the equatorward advance of ice sheets with increasing cold, the most obvious explanation of low-latitude glaciation in late Precambrian time is that the Earth was then glaciated from pole to pole during a very severe ice age. Such a situation might arise if the Sun's radiation were then anomalously low. This model of global glaciation, however, is inconsistent with numerous observations. High latitudes, being the spreading centres of glaciation, would be affected by cold climate for significantly longer intervals of time than low latitudes. The lack of direct evidence for high-latitude glaciation in late Precambrian time therefore militates against global glaciation. The lack cannot be explained by an absence of polar land masses, as certain areas did occupy high latitudes during the late Precambrian yet afford no evidence of glaciation [12]. Furthermore, global glaciation would not appreciably affect seasonality, and hence strong sea-

sonal changes of temperature would not be experienced in low latitudes. In fact there is much evidence for marked seasonal changes of temperature in low palaeolatitudes during late Precambrian glaciation, as indicated by the widespread occurrence of varvites [3,25,26] and periglacial features including sand wedges [17,49]. The known continuity of life through late Precambrian time may present a further objection to the concept of global glaciation. Finally, with regard to solar energy, it would be very difficult to unfreeze a frozen-over Earth, a 30% increase in solar luminosity being required by climate models [50]. Evidence that sunspot cyclicity similar to that of today operated in late Precambrian glacial time [25–27] suggests, rather, that the behaviour of the late Precambrian Sun was essentially "normal".

We are thus left with two possible explanations for the apparent distribution of very cold climate during the late Precambrian: (1) the axial geocentric dipole model of the Earth's magnetic field [51] is invalid for that time interval; and (2) the obliquity of the ecliptic was considerably increased ($> 54°$) during late Precambrian glaciation [3]. As discussed by McElhinny [51], ultimate testing of the validity of the axial geocentric dipole model over geological time appeals to palaeoclimatic evidence. The model is strongly supported by geological observation for the Phanerozoic, and hence it remains the cornerstone of palaeomagnetic interpretation. Any deviation from this model during the late Precambrian could not have been a short-lived event, as the pole obtained here for the Elatina Formation falls within the group of poles for the quasi-static interval 700–560 m.y. [46]. Invalidation of the axial geocentric dipole model for such a major interval of time would call into question other Precambrian palaeomagnetic data for which unequivocal support from palaeoclimatic evidence was lacking.

Alternatively, the climatic anomaly presented by low-latitude glaciation may be resolved by postulating a considerably increased obliquity of the ecliptic during late Precambrian glaciation. As discussed by Williams [3], substantial increase in obliquity would result in important changes in global climate:

(1) The ratio of radiation received annually at either pole to that received at the equator would be increased. When the obliquity exceeded 54° low latitudes would receive less insolation annually than high latitudes and hence would be preferentially glaciated.

(2) Seasonality would be greatly intensified, thus causing significant seasonal changes of temperature to extend into low latitudes.

(3) Climatic zonation would be weakened. As local climates would be delicately balanced, any Milankovitch-type fluctuations in insolation arising from short-term (10^4 to 10^6 years) changes in orbital parameters might cause exaggerated or abrupt changes of local climate. These changes could result in the close association of tillites with warm-climate indicators such as red beds and primary dolostones, which is in fact a puzzling feature of late Precambrian glaciogenic sequences [3,52].

It has been argued by Vanyo and Awramik [53], for non-glacial rocks ~ 200 m.y. older than the Elatina Formation, that late Precambrian palaeo-obliquity was little different from the present obliquity. They base their tentative conclusion on the growth behaviour of inclined columnar stromatolites. In most cases the causes of such inclination are unknown or can only be tentatively inferred. In some modern occurrences, however, current direction [54] and prevailing wind direction [55] appear to strongly influence the inclination of columnar stromatolites. Hence, until a widely accepted, satisfactory basis for interpretation of the stromatolite data is resolved, the claim of Vanyo and Awramik [53] should be viewed as highly speculative.

The predicted effects of global glaciation, glaciation for an obliquity of the ecliptic exceeding 54°, and glaciation for a non-axial geomagnetic field are compared in Fig. 9 Geological observations as a whole offer little support for the global-glaciation model. Of the other two possible explanations, the high-obliquity model appears favoured because of the widespread occurrence of late Precambrian glaciogenic rocks and their close association with apparent warm-climate indicators. Whichever explanation one prefers, it is evident that the occurrence of very cold climate near sea level in low palaeolatitudes during late Precambrian time joins palaeomagnetism and celestial mechanics in an enigma that calls for intensive research.

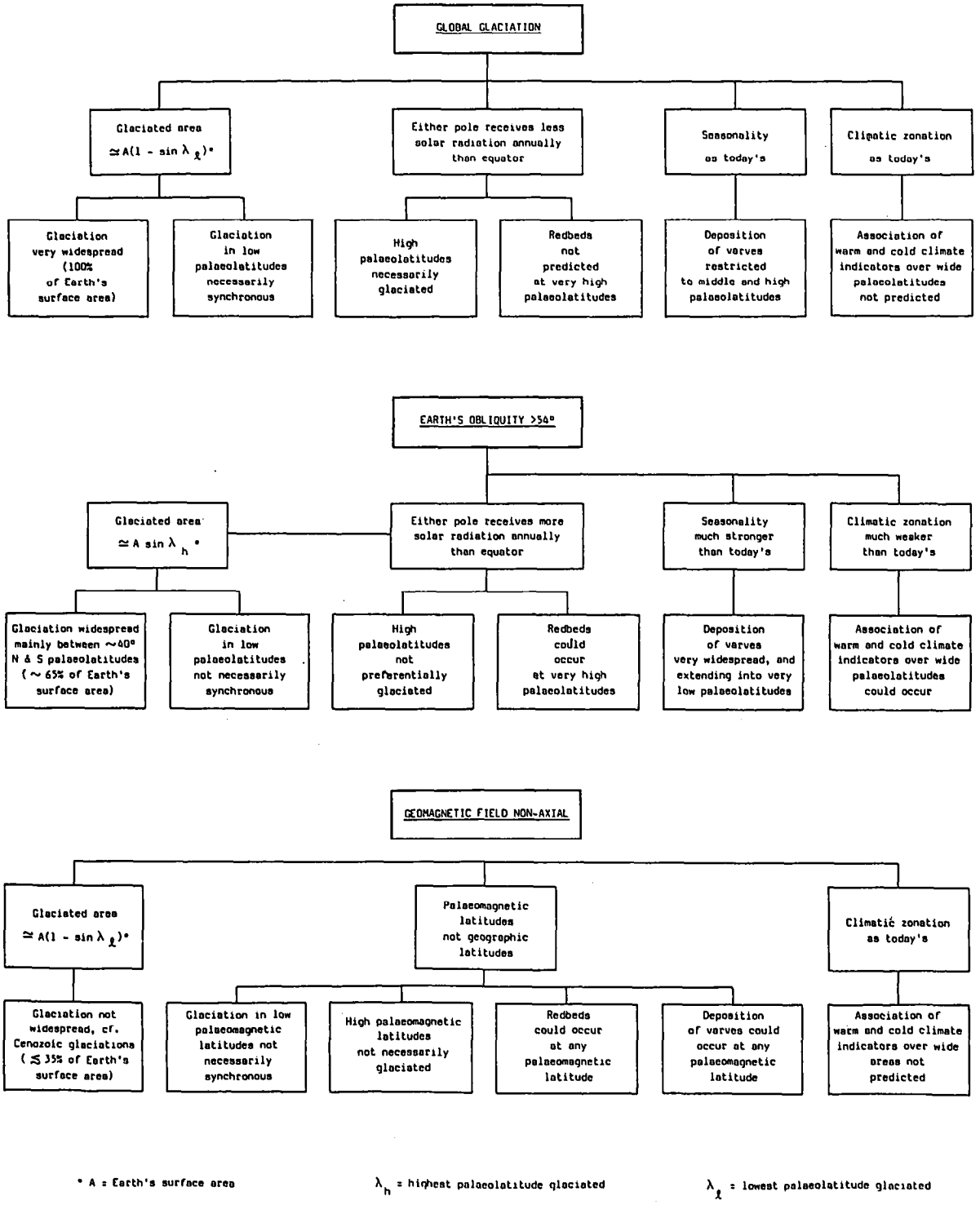


Fig. 9. Predicted effects of global glaciation, glaciation for an obliquity of the ecliptic $> 54°$, and glaciation for a non-axial geomagnetic field. Modified after Williams [56].

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